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Velocity Model Uncertainty Estimation in Area with No Well Calibration in Central Oman

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Abstract

Accurate depth conversion is a critical component in hydrocarbon exploration, as it directly influences the success of Trap definition and subsequently volume and first well exploration placement. This process involves converting seismic data recorded in time domain into depth domain using velocity model. Hence, the reliability of depth conversion is highly impacted by the velocity model uncertainty and the availability of well data. Small changes or uncertainties in the velocity model can significantly impact the accuracy of converting seismic data from time to depth. Conversion uncertainty increases with target depth; as the target becomes deeper, the associated uncertainty tends to grow. In addition, implementing depth conversion in regions characterized by low relief structure is highly sensitive to the accuracy of the velocity model. Due to the subtle vertical separations inherent in such settings. Therefore, developing a geologically consistent and data-constrained velocity model is critical to minimize uncertainty in depth conversion and ensure accurate assessments of volume estimation to ensure success in well placement. This paper demonstrates a case study from an area in central Oman, where the subsurface team attempted to build a robust velocity model estimation approach by calibrating a regional velocity model with Pre-stack Depth Migration (PSDM) velocity model. A key challenge in this project is lack of well data in area of interest as this work was implemented to plan the first well in the area. The primary target of this well is at depths of more than 5 km hence the uncertainty is higher

The main aim is to address the velocity model uncertainty by utilizing both the localized seismic Pre-stack Depth Migration (PSDM) velocity with regional velocity model trends generated using available wells from nearby fields. The team evaluated and compared both models and assured their validity to be used for structure map depth conversion. Hence, this outcome feeds into the volumetric estimation of subsurface trapped hydrocarbons. While both models confirm the presence of the trap, some variation between the models observed and was used to estimate uncertainty in volume calculation

Introduction

Velocity models are the backbone of subsurface imaging and reservoir characterization in the oil and gas sector. These models are essential for converting seismic travel times into true depth images, enabling geoscientists to identify hydrocarbon reservoirs with greater precision. To estimate the velocity distribution

within subsurface, it requires an integration between seismic data and well logs data aligned with geologic understanding. Well data and seismic data serve complementary roles in velocity prediction during oil and gas exploration. Well data provides high-resolution, depth-based measurements at specific locations, making it ideal for calibrating and validating models. In contrast, seismic data offers broader spatial coverage but is recorded in time, requiring conversion to depth using velocity models. While seismic surveys give a wide subsurface view, only well data can supply the precise "truth points" needed to refine and verify those models. Together, they create a more accurate and reliable picture of the Earth's interior. The availability of well data significantly strengthens velocity model reliability, serving as a vital constraint for calibrating seismic velocity and validating modeled depth conversions. Well logs, checkshots, and vertical seismic profiles (VSPs) provide ground truth measurements that anchor the model in actual subsurface conditions, minimizing uncertainties and increasing confidence in exploration decisions. Unfortunately, the AOI is undrilled and the closest drilled wells to target level are about 22 km far from the concept. An applied case study is provided to exemplify the method of velocity model estimation within exploration context, aimed at evaluating untapped reservoirs of Cambrian age. This process was performed during the prospect maturation and considers two Cambrian reservoirs as the primary targets in the value case. Upper reservoir (UR) deposited during in the latest Cambrian to the earliest Ordovician in fluvial and shoreface depositional environment. It consists of siltstones and claystone interbedded with amalgamated sandstone with an average porosity of approx. 7% and permeability of 0.56 mD. The thickness of the reservoir can reach up to several hundred meters and get thinning in the northern direction as the deposit moves from continental environment into offshore mud. It is sealed by a thick lateral extensive interval of micaceous, red-brown mudstones. The middle reservoir (MR) deposited during the middle Cambrian age in an alluvial and inland Sabkha to lacustrine environment with minor aeolian influence. It consists of micaceous shales intercalated with very fine- to fine- grained siltstones and sandstones. A sand-rich layer approximately 100 meters thick, situated in the middle of MR, constitutes a low- permeability Hydrocarbon reservoir. While the upper member of this reservoir acts as a seal as it is mainly made up of shales and siltstone. Besides that, this project includes a secondary objective such as the lower Cambrian reservoir (LR) that consists of thick sandstone of alluvial and aeolian deposition environment, as well as the Permian sandstone reservoir. The structural features are characterized by gentle, faulted dip-closures, and in some areas, include a stratigraphic pinch-out element. The low relief structures identified in TWT are very sensitive to depth conversion and with the absence of well control in the AOI; detailed velocity modeling is key to assessing the size of the price.

Relevant Data

The area of interest covered by new acquired 3D seismic data to unlock the potential of Cambrian reservoirs. As this area was previously evaluated using 2D seismic data that was able to identify clusters of 4-way dip closure leads with low relief structural component. The new 3D data confirmed the presence of these clusters with slight changes in geometry and size.

Pre-stack depth migration (PSDM) velocity model was also received from the processing company along with the new seismic. Hence it was used for the depth conversion procedure. Moreover, a regional velocity model was built using all well data (nearest is 22km away) with layered velocity approach is used as another model to convert the seismic. The AOI sets to the south of the drilled wells and the regional model over the leads will be using mapped surfaces and extrapolated velocity functions.

In terms of well data, which is considered the most confident data point for depth conversion, unfortunately no wells are available in the area. However, a shallow well (SX) towards the south direction is available and can be used to calibrate both velocity models at shallow depth only (the well did not go as deep as the target level of the mapped concepts). Hence, uncertainty remains. The distance between this well and the mapped concepts is around 19 km. [Fig1](#). Shows the velocity model coverage and area of interest.

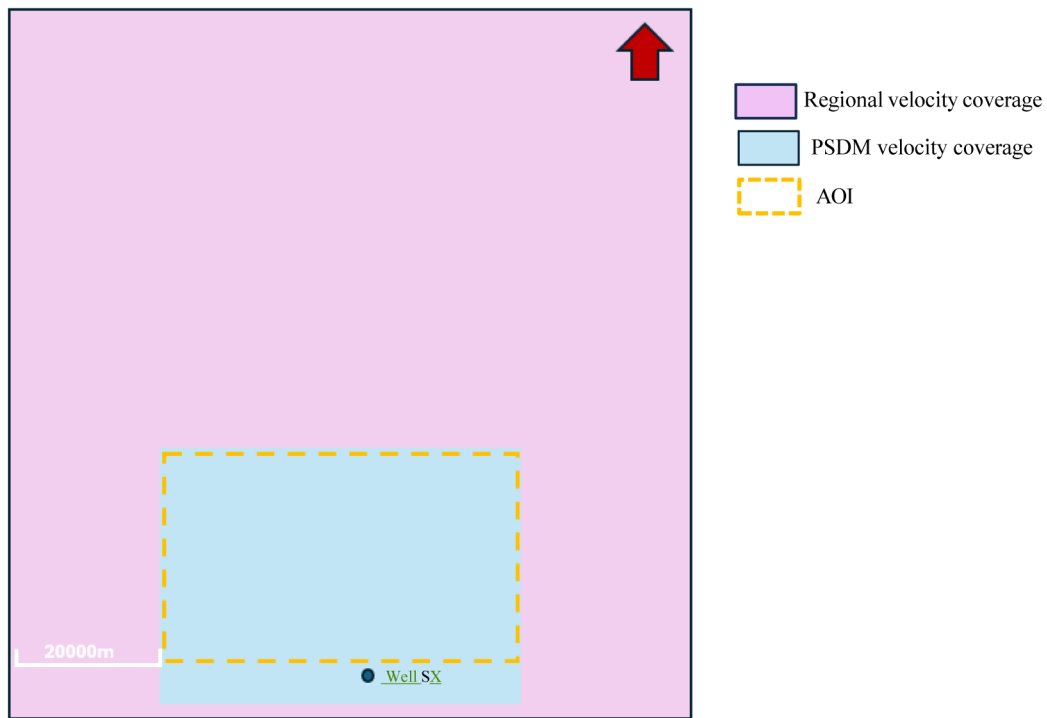


Figure 1—display the two velocity models and the corresponding focused area, where the regional model

Workflow

In oil and gas Industry, there is an essential workflow processes to be followed in order to mature a commercial prospect as illustrated in [Chart. 1](#). These steps are dependently related hence, the precision in the outcome of each step can directly impact the implementation of the next one. In this piece of work the focus will be on the velocity model building, time depth conversion and volumetric calculation.

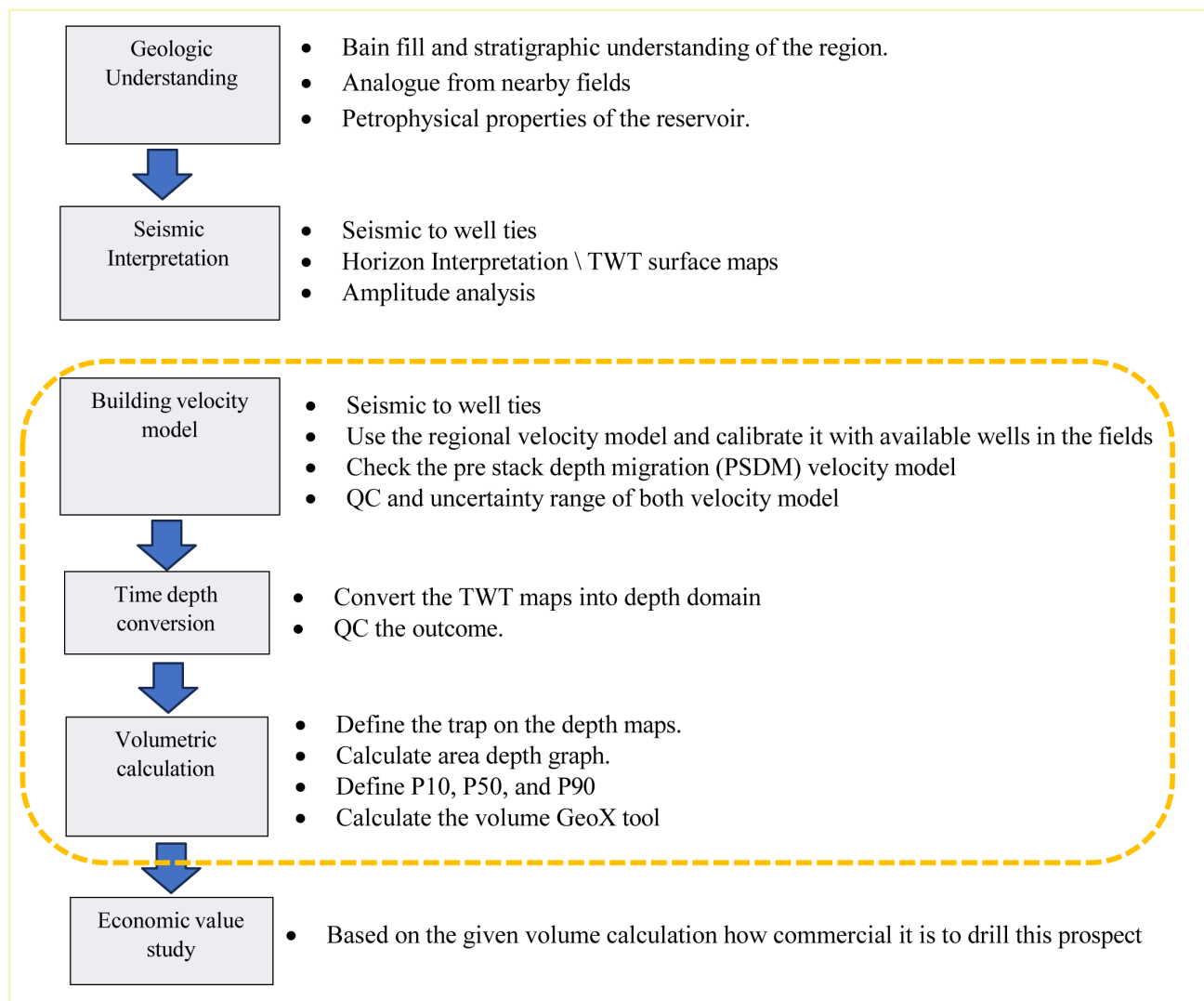


Chart 1—explains the prospect maturation process in oil and gas industry.

Methodology

To address the depth accuracy due to lack of depth calibration from wells data, two independent velocity models from different data inputs were utilized by a team consisting of geologists, geophysicists and processing geophysical technical authority. First, the team used a regional velocity model that is extended to the AOI and corrected to a shallow well control (wells) south of AOI. Since the aim is to build a hybrid model that also captures the granularity of the recently processed PSDM data; they used such PSDM velocity model that indicates similar trends as the regional model. The opportunity in this area represents a cluster of low relief structure, however, both velocity models show similar structures with variable sizes and relief. To Understand the quality of the PSDM model, the team investigated the raw migrated seismic gathers. This was a crucial step to examine the quality of seismic data, moveout correction analysis, and structure uncertainty. The raw gathers reveal a consistent depth positioning of the reflectors across different offsets which indicate the reliability of PSDM velocity model.

For the assessment of depth sensitivity of the structure and volumetrics, the team compared the depth conversion results of the two models, and they decided to use the PSDM model for final depth conversion since it provides extra details. However, for volumetric calculation, they consider both models and assume base case scenario (P50) to be the midpoint between the two models.

Velocity model evaluation

The regional velocity model covers the area of interest and calibrated using wells from blocks northern side of the AOI. The only calibration solid point in the south is SX well does not reach the target reservoir of this project. Therefore, implementing depth conversion using regional velocity model only will not capture all the uncertainty.

The regional velocity model was built using three main velocities input which are: X20 well- based velocity cube, seismic velocity volume, and Pseudo velocity points driven from time and depth pair at each well location. Pseudo velocity interval (Vint) considered as hard data during building the velocity maps and merged with volume velocities which guide the model. Also, four key horizons are interpreted in time domain as input to build the regional model. Fig1. Illustrate the methodology to produce Vint points at a well location at each interval, where the TWT thickness obtained from picked horizons and the depth thickness from the well tops isochore for each layer. Then extract the interval velocity for each layer using the TWT horizons, X20 velocity volume and seismic velocity volume. The resulting velocity maps were smoothed to capture the regional trends with no high frequency changes. To assess the velocity maps, the team investigated the relation between the Vint points and TWT thickness maps, X20 Vint maps, and seismic volume Vint. However, the seismic velocity volume used to build the regional velocity model is different than PSDM volume that is discussed in this paper.

The other velocity model used in this study is PSDM associated with seismic c [ube. To quality control of the PSDM velocity model, the team address the raw seismic gathers. Looking at the gathers from different offset to check how consistent is the seismic reflector at target level. This provides the team with key information about how accurate the moveout correction is applied to the gathers. In which, moveout correction is directly relied on the velocity's assumption of subsurface and whether it truly represents the earth. In this project the raw gathers found to be flat and consistent at target level which means the movement correction applied to the seismic during processing is aligned with subsurface and the seismic waves are accurately migrated to their true location as shown in Fig. 4. This ensures the validity of PSDM model and reduces the uncertainty in seismic imaging of subsurface features.

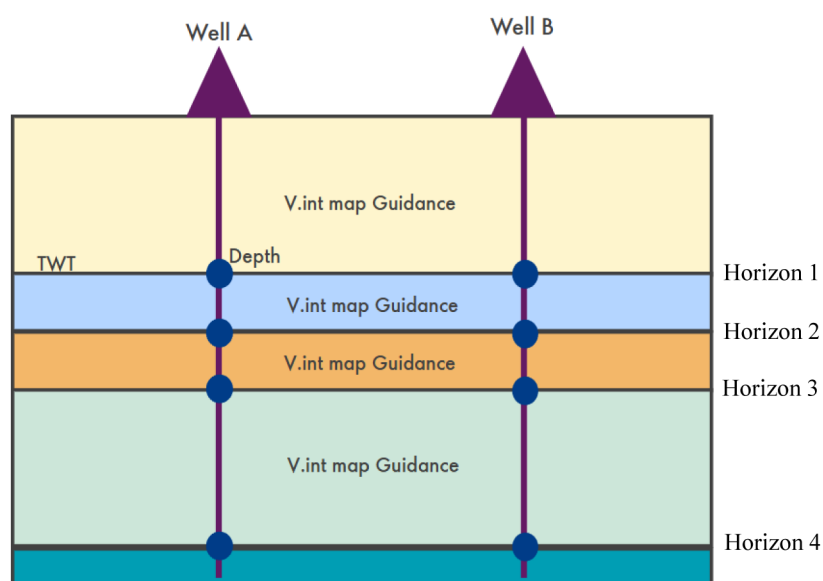


Figure 2—display the Velocity interval calculation at a well location utilizing the key TWT horizons with the corresponding well tops.

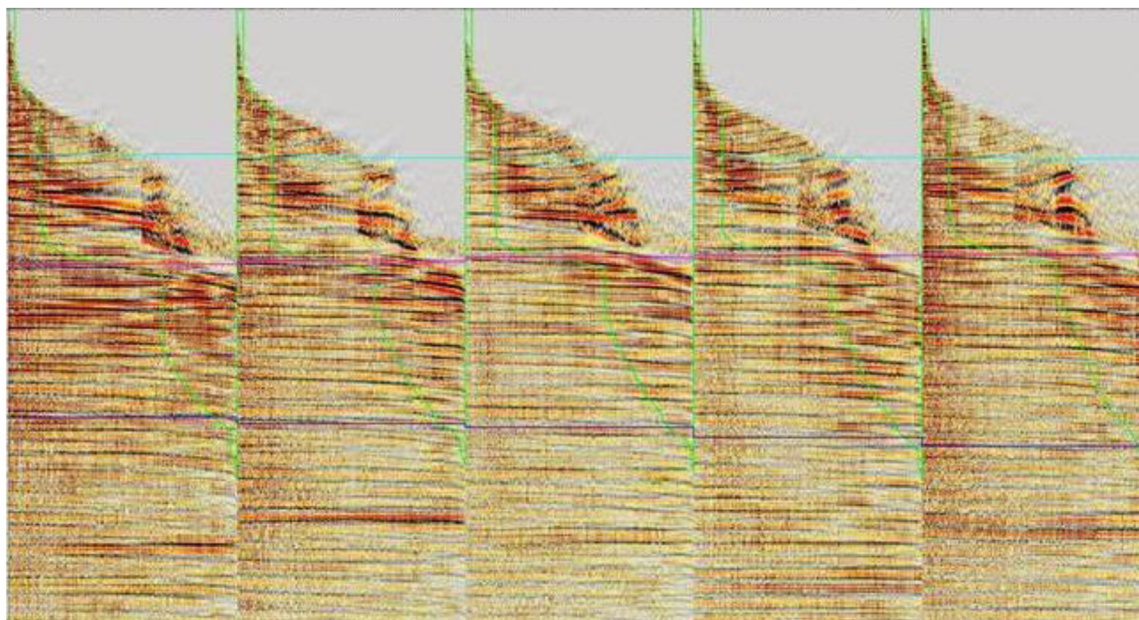


Figure 3—represent the raw migration gathers which shows that the reflectors are flat at target level at different offset.

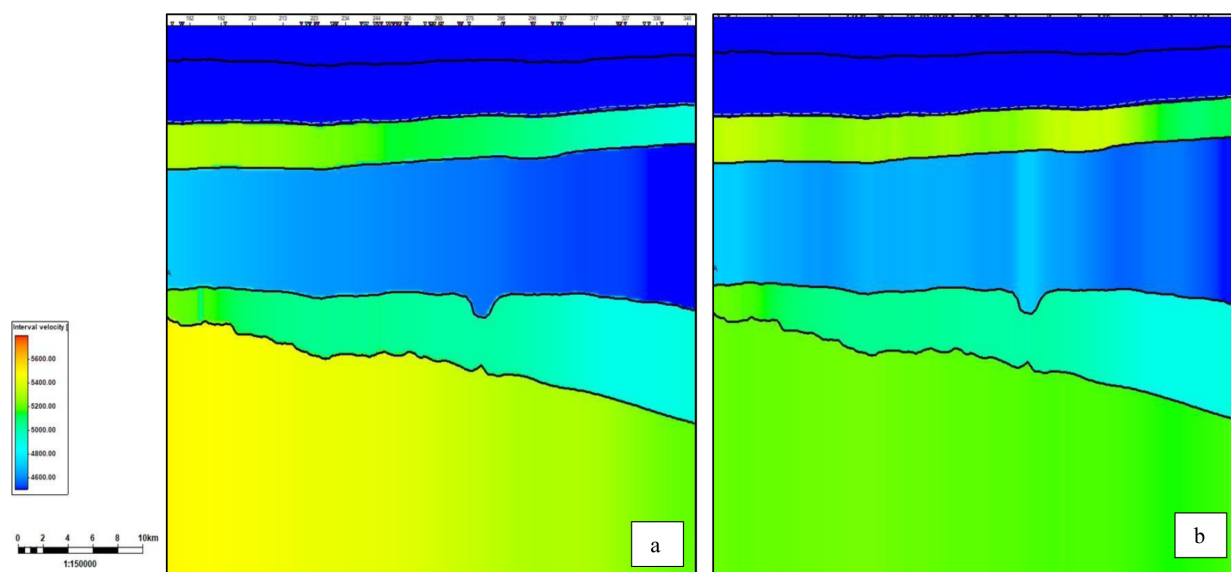


Figure 4—display the velocity model in velocity domain with the key horizons in this area, where (a) represents regional velocity model; and (b) PSDM velocity model.

For this study both velocity models were used to depth convert the reservoir time surface map into depth map to support trap definition and volumetric calculation and then comparing the output. Assessing the velocity models in velocity domain, the similarity between the two velocity models trends can be observed, however PSDM model provides extra details as shown in Fig. 4.

Depth Conversion

The prospect structure exists in both maps with some changes in area as shown in Fig. 5. However, as those structures consider having low relief hence velocity accuracy can have a high impact and the uncertainty on the areal extent of the relief remains due to the lack of hard data calibration in the AOI.

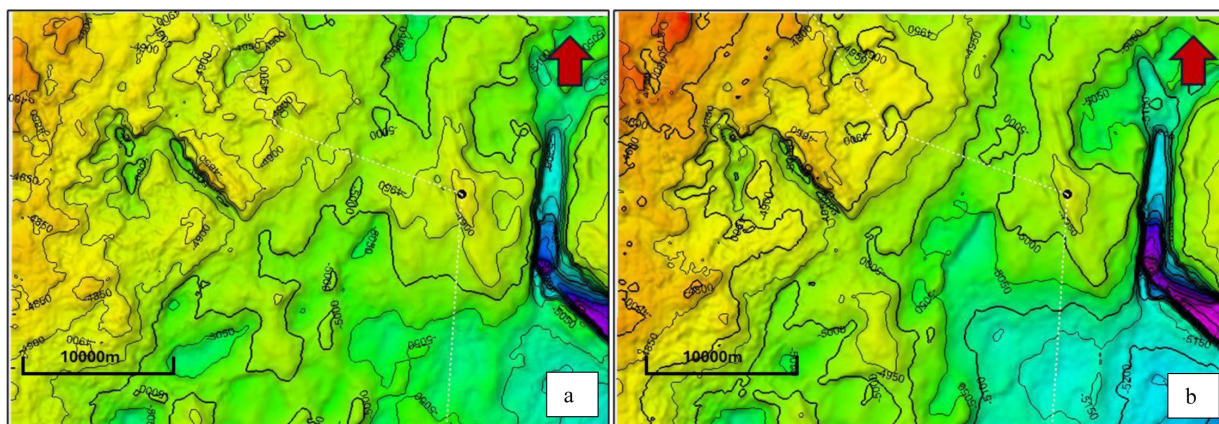


Figure 5—Illustrate the depth map of the top reservoir converted from time domain using (a) regional velocity model; (b) Pre stack depth migration model.

Moreover, the two depth surfaces displayed on seismic section to address the differences as represented by Fig. 6. The target appeared to be deeper in the surface converted using PSDM model than the regional model. Another observation found in PSDM model is a general flexing in NW – SE direction. One of the uncertainties in PSDM velocity model, is lack of deeper control from well data.

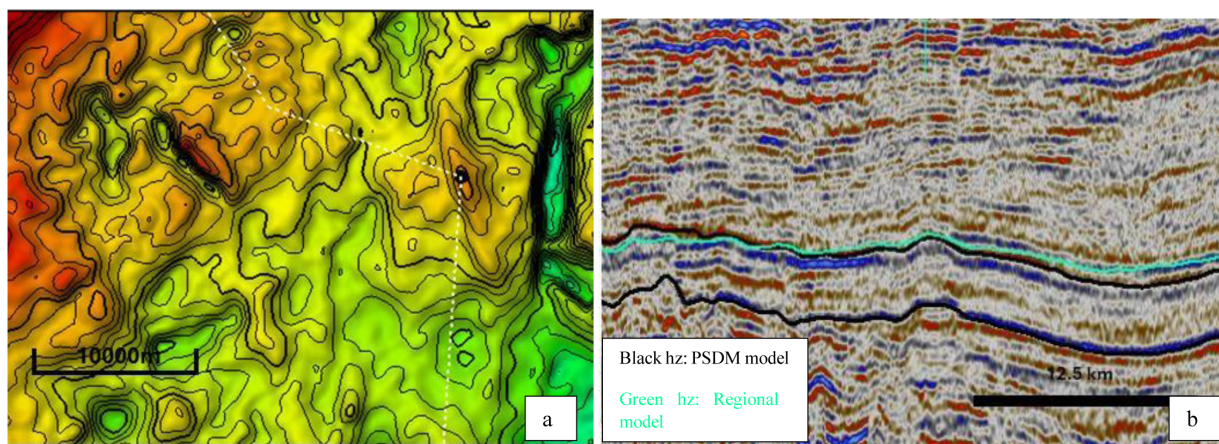


Figure 6—A section across the prospect structure from S – NW direction: (a) map view; (b) seismic section view to show a comparison between regional velocity model horizon and pre-stack depth migration (PSDM) model horizon.

Although the primary target in this project is deeper reservoirs, the team also identified secondary shallower reservoir. Hence It was important to check the velocity model effects at shallower surfaces. Fig. 7 – shows the depth map at shallower surface where the trap assumption is channel sands crossing structural noses. Comparing the two surfaces reveal that the structure noses in the map converted using PSDM is narrower than the regional velocity. This areal difference has an impact on area depth graph calculation that directly impacts the volume of potential Hydrocarbon.

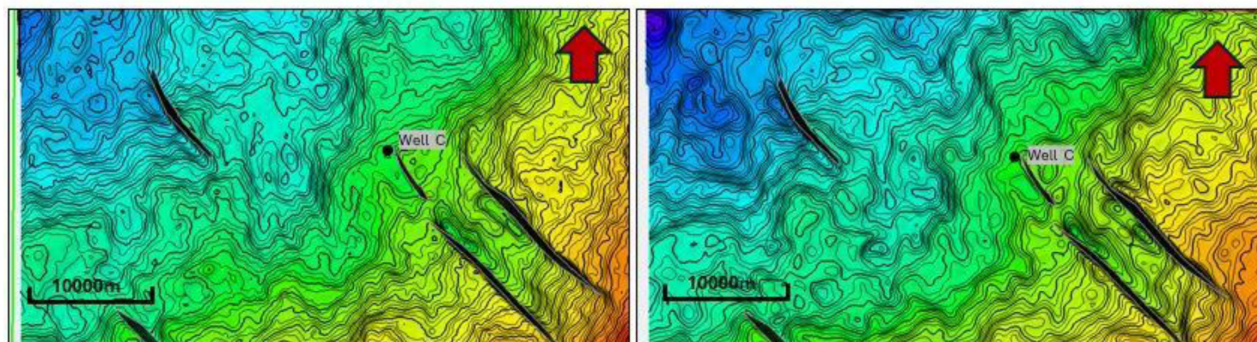


Figure 7—shows depth maps for a secondary shallower target in the area converted using different velocity model to address structure differences, where (a) converted using PSDM velocity model; (b) with regional velocity model.

In addition, A comparison between the PSDM model and regional model was carried out at all the key horizons surface as illustrated to in Fig. 8, to understand the depth difference between the two models. In general, shallower surfaces seems to be at similar depth in both velocity models, but some difference is observed in the deeper surface.

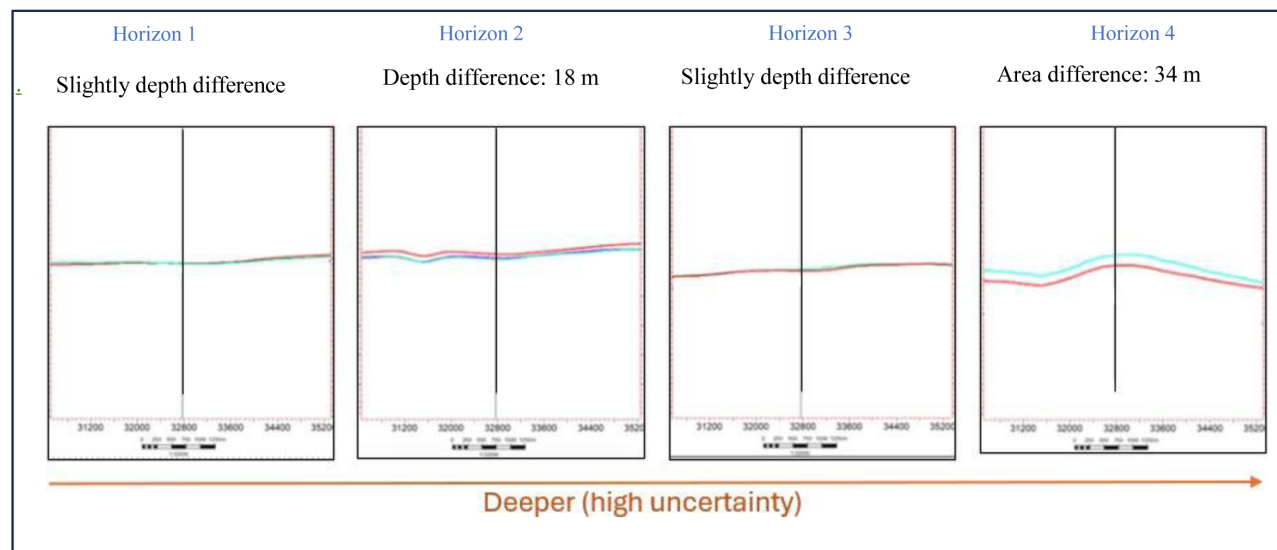


Figure 8—shows four key surfaces that represent velocity breaks in the area converted using regional model (green line) and PSDM model (red line) with the difference between the two models.

Trap Definition

Volume represents a three-dimensional measurement defined by length, width and depth. Accurate depth conversion implementation has a vital role to estimate the true vertical extent of the reservoir, otherwise that might cause significant over or underestimation of subsurface volume. Many economic decisions rely on volume estimation such as well placement, number of well, production forecast, and investment decisions. Although the final depth maps were converted using the PSDM model, for area depth graph both models were used to capture all the uncertainty. Fig. 9 – represent prospect outline comparison between the two independent models. The area difference between the two models reaches 30%. Hence, the team decided to consider Regional and PSDM models as P30-P70 cases respectively and base case scenario (P50) is the midpoint between the two. And use a correction factor to calculate high case (P10) and low case (P90) scenarios. This approach ensures that the uncertainty in velocity model is considered in volume estimation. Moreover, the same methodology in volume calculation was used to evaluate other prospects in this area.

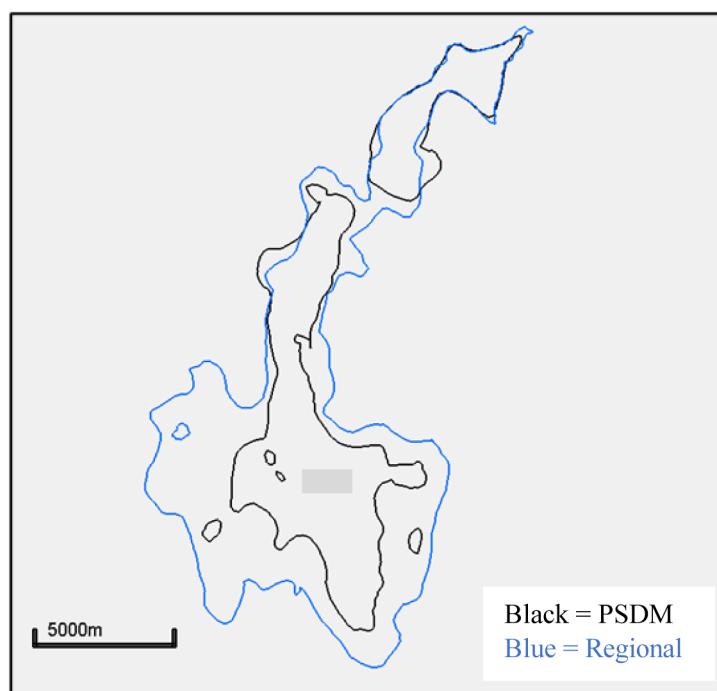


Figure 9—Shows the difference in area structure between regional model (blue) and PSDM (Black).

Conclusion

In prospect maturation, building a precision velocity model and accurate depth conversion is fundamental to ensure producing reliable subsurface interpretation. The velocity model controls the way seismic energy travels through time and then converted into true depth, this process shapes the geometry and vertical extent of subsurface layers. Errors in this transformation can misplace subsurface structural features, distort the target's spatial boundaries, and lead to error in thickness estimates— which in turn directly impact volumetric calculations. To reduce the uncertainty in velocity modeling, a solid data is required to calibrate the model. Unfortunately, in the exploration stage this could be a challenge due to lack of wells. In this project the team implements a new approach to handle this uncertainty and combines two velocity models for volume calculation. Each model was tested individually and evaluated to ensure their reliability.

Pre stack depth migration (PSDM) model was corrected using a shallow well (S), however it has some limitation due to lack of deeper control. Investigating the raw migration gathers proved the reliability of this model as the seismic reflectors seems to be consistent in all offsets. The reservoir structure is found to be deeper in this model compared to regional velocity model with a smaller areal extent of the structure. Moreover, a general flexing in SE - NW direction is observed in this model hence structure relief uncertainty remained.

Regional model corrected includes all wells from nearby fields northern the area of interest and corrected to a shallower well (SX) in the southern direction of AOI. Although it's showing the same structure as the PSDM model, however, it provides less details.

To conclude, both models show similar trends and hence both can be used for depth conversion. Although the structure remains in both models, there is a difference in area and the depth of the reservoir. Considering only one velocity model can lead to misestimation of reservoir resources. Taking this into account, both velocity models were used for volumetrics calculation in order to reduce the uncertainty.

The outcomes of this investigation have successfully guided the selection of optimal well location to test the structure.

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